

C_p -SEMI-EMBEDDINGS ALWAYS EMBED

AGENT LANE 05

SOURCE AND TARGET

The source paper is Kakol–Kurka–Sliwa, *Lotz-Peck-Porta and Rosenthal's theorems for spaces $C_p(X)$* , arXiv:2509.08634 [1]. On page 4, Problem 10 asks:

Let X be an infinite compact space. Is it true that each semi-embedding $T : C_p(X) \rightarrow C_p(Y)$ is an embedding for any compact space Y if and only if X is scattered?

A more specific version of Problem 8 might be also the following

Problem 10. Let X be an infinite compact space. Is it true that each semi-embedding $T : C_p(X) \rightarrow C_p(Y)$ is an embedding for any compact space Y if and only if X is scattered.

Note that if $T : C_p(X) \rightarrow C_p(Y)$ is embedding with both X , Y compact and Y is scattered, then X is scattered; this follows from [1, Theorem III.1.2]. Clearly, the converse implication fails in general. Having in mind Problem 10 we prove the following special case; this will follow from our Corollary 24.

Here $C_p(X)$ denotes $C(X)$ with the topology of pointwise convergence. Following the source paper, a continuous linear map $T : E \rightarrow F$ between locally convex spaces is *polar* if there is a base $\mathcal{U}$ of closed zero-neighborhoods in E such that $T(U)$ is closed in F for every $U \in \mathcal{U}$. A *semi-embedding* is an injective polar map.

RESULT

Problem 10 has a negative answer. The proposed “if and only if X is scattered” characterization is false, because the semi-embedding property does not detect scatteredness at all: between C_p -spaces, every semi-embedding is automatically a topological embedding.

PROOF INTUITION

A continuous linear map $T : C_p(X) \rightarrow C_p(Y)$ is controlled point by point on the codomain. For each $y \in Y$, the functional $\delta_y \circ T$ on $C_p(X)$ depends on only finitely many points of X . These finite-support functionals generate the topology inherited by the range $T(C_p(X))$. If T were not an embedding, some point evaluation δ_x on X would not be continuous for this inherited topology. The polar condition rules this out. Indeed, take a polar

neighborhood whose preimage controls $f(x)$. It contains a finite-coordinate kernel $M = \{f : f|_F = 0\}$, and its closedness in the inherited topology forces the closure of M to remain inside the strip $|f(x)| \leq 1$. Since that closure is a linear subspace, δ_x annihilates it. The annihilator of that closure is exactly the part of the inherited dual supported on the finite set F , so δ_x must already belong to the inherited dual.

Theorem 1. *Let X and Y be Tychonoff spaces. Every semi-embedding*

$$T : C_p(X) \longrightarrow C_p(Y)$$

is a topological embedding.

Proof. Let $E = C(X)$ as a vector space, and let $\tau_p = \sigma(E, \{\delta_x : x \in X\})$ be the topology of $C_p(X)$. For each $y \in Y$, put

$$h_y = \delta_y \circ T \in E'.$$

Since every continuous linear functional on $C_p(X)$ is a finite linear combination of point evaluations, each h_y belongs to $\text{span}\{\delta_x : x \in X\}$. Let

$$H = \text{span}\{h_y : y \in Y\} \subset E'$$

and let $\rho = \sigma(E, H)$. The topology ρ is precisely the topology pulled back from the subspace $T(E) \subset C_p(Y)$. Since T is injective, $T : (E, \rho) \rightarrow T(E)$ is a linear homeomorphism onto its range. Also $\rho \leq \tau_p$, because $T : C_p(X) \rightarrow C_p(Y)$ is continuous.

We claim that $\delta_x \in H$ for every $x \in X$. Fix $x_0 \in X$. Because T is polar, there is a base $\mathcal{U}$ of closed τ_p -neighborhoods such that $T(U)$ is closed in $C_p(Y)$ for every $U \in \mathcal{U}$. Choose $U \in \mathcal{U}$ with

$$U \subseteq \{f \in E : |f(x_0)| \leq 1\}.$$

The set $T(U)$ is closed in $C_p(Y)$, hence closed in the range $T(E)$. Transporting this statement through the homeomorphism $(E, \rho) \cong T(E)$, we get that U is ρ -closed.

Since U is a τ_p -neighborhood of zero, there are a finite set $F \subset X$ and $\varepsilon > 0$ such that

$$V = \{f \in E : |f(t)| < \varepsilon \text{ for all } t \in F\} \subseteq U.$$

Let

$$M = \{f \in E : f(t) = 0 \text{ for all } t \in F\}.$$

Then $M \subset V \subset U$. Since U is ρ -closed, the ρ -closure $\overline{M}^\rho$ is contained in U . The set $\overline{M}^\rho$ is a linear subspace. Therefore, if $g \in \overline{M}^\rho$, then $ag \in \overline{M}^\rho \subset U$ for every scalar a . As $U \subseteq \{|f(x_0)| \leq 1\}$, this implies $|ag(x_0)| \leq 1$ for every scalar a , hence $g(x_0) = 0$. Thus

$$\delta_{x_0} \in (\overline{M}^\rho)^\perp.$$

It remains to identify this annihilator. In the weak topology ρ , the closure of M is

$$\overline{M}^\rho = \bigcap_{\lambda \in H \cap M^\perp} \ker \lambda.$$

The annihilator of this intersection is the pointwise closure of $H \cap M^\perp$ in the algebraic dual. But

$$M^\perp = \text{span}\{\delta_t : t \in F\}$$

is finite-dimensional, so the subspace $H \cap M^\perp$ is pointwise closed. Hence

$$(\overline{M^\rho})^\perp = H \cap M^\perp$$

as subspaces of the algebraic dual. Consequently $\delta_{x_0} \in H$. Since x_0 was arbitrary, every point evaluation δ_x is ρ -continuous.

The topology τ_p is generated by the point evaluations δ_x . Thus $\tau_p \leq \rho$. We already had $\rho \leq \tau_p$, so $\rho = \tau_p$. Therefore the range topology inherited from $C_p(Y)$ is exactly the original topology of $C_p(X)$, and T is a topological embedding. $\square$

Corollary 2. *The characterization proposed in Problem 10 of [1] is false.*

Proof. Take $X = [0, 1]$. This is an infinite compact non-scattered space. By the theorem, for every compact space Y , each semi-embedding $T : C_p([0, 1]) \rightarrow C_p(Y)$ is an embedding. Thus the property appearing in Problem 10 holds although X is not scattered. $\square$

VERIFICATION NOTES

No computational check is involved. The proof uses only:

- the standard description of $C_p(X)'$ as finite linear combinations of point evaluations;
- the fact that $T(U)$ closed in $C_p(Y)$ makes U closed for the range-induced topology $\rho = \sigma(E, H)$;
- the closure/annihilator formula for a subspace in a weak topology.

The argument also explains why the dense-restriction map mentioned in the source paper is not a semi-embedding: the induced weaker topology does not make the missing point evaluations continuous, so it cannot have the required closed-neighborhood base.

NOVELTY AND SEARCH BOUNDS

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2509.08634, the title keywords, and the core phrases “semi-embedding”, “ $C_p(X)$ ”, “ $C_p(Y)$ ”, and “polar map”. No existing packet for this result was found.

External web searches on 2026-06-15 used exact and near-exact phrases from Problem 10, including “Let X be an infinite compact space semi-embedding $C_p(X) C_p(Y)$ ”, “semi-embedding $C_p(X) C_p(Y)$ compact scattered”, and “polar map $C_p(X)$ quasibarrelled semi-embedding”. No later paper explicitly answering Problem 10 was found. The result should nevertheless be reviewed with medium novelty confidence, because the proof is short and may exist as folklore in C_p -theory.

HUMAN REVIEW RECOMMENDATION

Review as a candidate full negative answer to Problem 10. The key checks are: first, that the source definition of semi-embedding allows the cofinal closed neighborhood U used above; second, that the induced topology on $T(E)$ is exactly $\sigma(E, H)$; and third, that the double-annihilator step is valid for the weak closure of M , using the finite-dimensionality of $M^\perp$.

REFERENCES

- [1] J. Kakol, O. Kurka, and W. Sliwa, *Lotz-Peck-Porta and Rosenthal's theorems for spaces $C_p(X)$* , arXiv:2509.08634, 2025.
- [2] A. V. Arkhangel'skii, *Topological Function Spaces*, Kluwer Academic Publishers, 1992.